

Sam Nguyen

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EDUCATION

Iowa State University

Bachelor of Science, Mechanical Engineering, 4.00 GPA

Expected Graduation: May 2029

Ames, Iowa

WORK EXPERIENCE

Thingz

May 2026 – Aug. 2026

Mechanical Engineering Intern

Galena, Illinois

- Designed and managed large electrical substations in Autodesk Inventor using point cloud data and engineering drawings to provide a digital twin representative of real-world equipment and environment
- Developed accurate and detailed models of electrical substation equipment, producing over 1,000 parts and sub-assemblies
- Rapidly created and managed a library of standardized substation parts and assemblies to streamline the CAD modeling process, reducing design turnaround time by roughly 70% and providing reference models for future projects
- Operated 360° LiDAR scanners and cameras to capture point cloud data across entire substations on-site
- Coordinated efforts within a cross-functional team of engineers, delivering weekly reports on modeling progress
- Consistently adhere to modeling principles and requirement standards to ensure proper integration with existing parts, assemblies, and naming conventions

EXTRACURRICULAR

Formula SAE Electric Vehicle (FSAE EV)

Sep. 2025 – Present

Suspension Subsystem Member

Iowa State University

- Independently designed and manufactured a static strain gauge calibrator, involving statics calculations, hands-on tooling experience, and welding, culminating in significant data acquisition for the team's R&D of strain gauges
- Created and implemented various 3D-printed jig parts to ensure the precise location of critical suspension component geometries during manufacturing such as the A-Arms, push rods, and tabs, on all 4 corners of the vehicle
- Oversee and own the bell crank/rocker assembly of the latest design cycle of the vehicle, utilizing design tables, proper modeling practices, and FEA analysis to execute and validate decisions while also improving knowledge transfer
- Collaborated with external design teams to ensure seamless integration between suspension parts in ARB assembly
- Analyzed bearing specs to determine the best solution for use case, considering packaging conflicts and costs
- Precisely 3D modeled parts to be easily scalable through use of parametric modeling techniques and
- Researched and designed various solutions to connect suspension blade components to the anti-roll bar

SolidWorks Club

Aug. 2025 – Present

Sophomore Representative

Iowa State University

- Achieved SolidWorks CSWP Certification through weekly CAD assignments and rigorous final 3-hour assessment which tested part modeling, design iteration, and assembly skills, scoring a flawless 100%
- Evaluated engineering drawings of models and assemblies in accordance with ASME Y14.5 and GD&T standards
- Elected as Sophomore Representative after first year of membership demonstrating rapid growth and mentorship
- Assisted club members in learning core SolidWorks skills by providing 1-on-1 feedback and first-hand experience
- Led group instructional sessions to foster a greater skill set among ISU engineers using computer-aided-design software
- Demonstrated excellent interpersonal communication skills to achieve a record of 163 new member sign-ups at info booth

PROJECTS

Fishing Pole Holder

Sep. 2025 – Nov. 2025

- Assessed design requirements for the product to ensure proper integration, using digital calipers and paper sketches
- Developed and revised product design using SolidWorks CAD models and a first principles engineering perspective
- Manufactured final design through use of FDM 3D printing on a Bambu Lab X1C, material choice being PETG filament

SKILLS AND CERTIFICATIONS

SolidWorks CSWP, Autodesk Inventor, 3D Printing (FDM), Python, MATLAB Onramp, Microsoft Office (Excel, Word, PowerPoint)